Semester One Examination, 2025

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section One: Calculator-free

If required by your examination administrator, please place your student identification label in this box

WA student number: In figures

In words

Circle your Teacher's Name: Mrs Bestall Mr Buckland Mrs Ducat Mrs Fraser-Jones Mr Kim Mr Lei Mr Luzuk Ms Narendranathan Mr Ng Mr Safa Mr Tanday

Time allowed for this section

Reading time before commencing work: five minutes
Working time: fifty minutes

Number of additional answer booklets used (if applicable):

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	12	12	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet preferably using a blue/black pen.
Do not use erasable or gel pens.

3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.

4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

5. It is recommended that you do not use pencil, except in diagrams.

6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
1	6	
2	8	
3	7	
4	7	
5	10	
6	7	
7	7	
Section 1 Total	52	
Weighted (×0.6731)	35%	
Section 2 Weighted	65%	
Total	100%	

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Section One: Calculator-free **35% (52 Marks)**

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1 **(6 marks)**

(a) Determine the equation of the straight line through the point $(2, -7)$ that is perpendicular to the line with equation $y = -\frac{1}{4}x + 1$. (2 marks)

(b) Function g is defined by $g(t) = 6t^7 - 3t^5 - 11t^4 - t^3 + 5t^2 - 2$. State

(i) the degree of g. (1 mark)

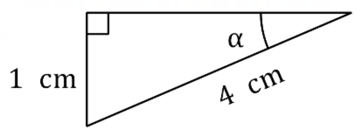
(ii) the coefficient of the t^5 term. (1 mark)

(c) Solve the equation $(x - 7)^2 = 3$. (2 marks)

Question 2 **(8 marks)**

(a) Express the angle $9°$ in radian measure. (1 mark)

(b) Determine the value of $\tan\alpha$ in the right triangle shown below. (2 marks)



(c) Solve the equation $\sin(2x) \times \cos\left(2x - \frac{\pi}{2}\right) = 1$ for $0 \leq x \leq \pi$. (5 marks)

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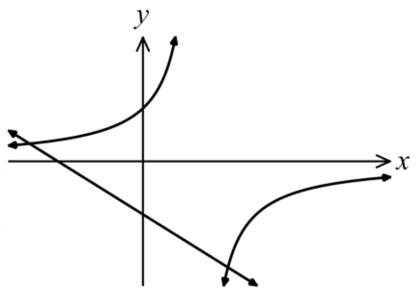
Question 3 **(7 marks)**

The graphs of the hyperbola $y = \dfrac{c}{2-x}$ and the straight line $x + y + 3 = 0$ are shown at right, where c is a constant. The hyperbola cuts the vertical axis when $y = 3$.



(a) Determine the slope of the straight line. (1 mark)

(b) Determine the value of c. (1 mark)

(c) State the equation of each asymptote of the hyperbola. (2 marks)

(d) Determine the x-coordinate of each point of intersection of the hyperbola and the straight line. (3 marks)

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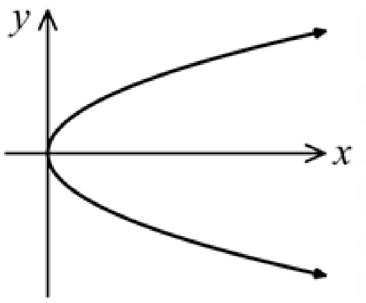
Question 4 **(7 marks)**

The relation $y^2 = x$ passes through the points $(a, 4)$ and $(9, b)$.

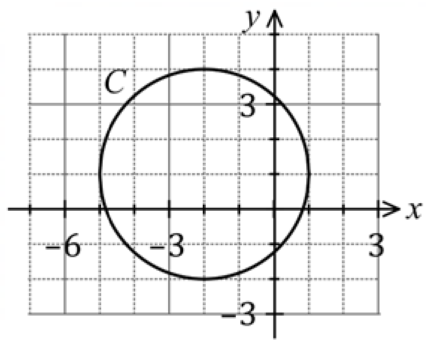


(a) State all possible values for a and b. (2 marks)

(b) Write down the name given to the shape of the graph of $y^2 = x$. (1 mark)

(c) State, with reasoning, whether the relation $y^2 = x$ is a function or not. (1 mark)

The graph of circle C is drawn on a cartesian plane.



(d) State the radius of circle C. (1 mark)

(e) Determine the equation of circle C. (2 marks)

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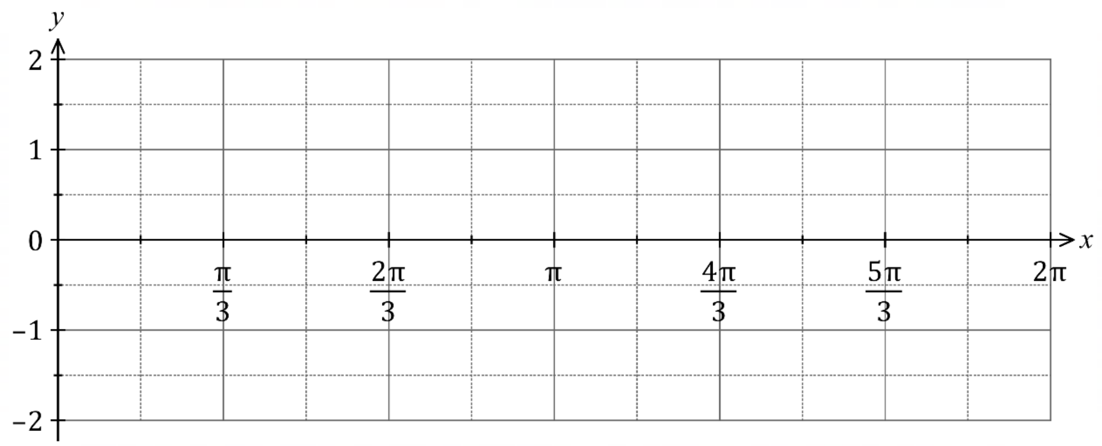
Question 5 **(10 marks)**

(a) State the amplitude and period of the function $f(t) = 2 - 5\sin(4t)$, where $0 \le t \le 20\pi$. (2 marks)

(b) Sketch the graph of $y = -2\cos\left(x - \frac{\pi}{3}\right)$ on the axes below for $0 \le x \le 2\pi$. (4 marks)



(c) Evaluate $\sin\left(\frac{\pi}{12}\right)$. (4 marks)

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Question 6 **(7 marks)**

The events A and B are such that $\mathrm{P}(A) = 0.4$ and $\mathrm{P}(B) = 0.5$.

(a) Determine $\mathrm{P}(A \cup B)$ when $\mathrm{P}(A \cap B) = 0.18$. (2 marks)

(b) Determine $\mathrm{P}(A \mid \bar{B}\,)$ when $\mathrm{P}(\bar{A} \cup B) = 0.92$. (2 marks)

(c) State, with justification, whether A and B are independent events when $\mathrm{P}(A \cap \bar{B}) = 0.15$. (3 marks)

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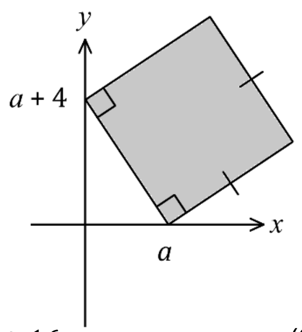
Question 7 **(7 marks)**

A computer screensaver shows a shaded square that has vertices at $(0, a+4)$ and $(a, 0)$.



Nick is an animator who is bored of staring at a static image. So, Nick creates a computer program that animates the square by changing the value of a, so that $-6 \leq a \leq 6$.
The image moves and resizes but remains a square.

(a) Show that the area of the square is given by $A = 2a^2 + 8a + 16$. (2 marks)

(b) Determine the area of the square when $a = -6$. (1 mark)

(c) Determine the coordinates of the turning point of the graph of $A = 2a^2 + 8a + 16$. (2 marks)

(d) Determine the minimum and maximum area of the square during the animation. (2 marks)

End of questions

Supplementary page

Question number: _________

Supplementary page

Question number: _________

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